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DIELECTRIC FIN STRUCTURES WITH VARYING HEIGHT

Abstract

A semiconductor device includes a semiconductor fin structure extending in a first direction on a substrate and a first dielectric fin structure extending parallel to the fin structure, the first dielectric fin structure being underneath a gate structure extending in a second direction that is perpendicular to the first direction. The device further includes a second dielectric fin structure extending parallel to the fin structure, the second dielectric feature being positioned beneath a gate cut feature. A top surface of the first dielectric fin structure is higher than a top surface of the second dielectric fin structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/384,092, titled “DIELECTRIC FIN STRUCTURES WITH VARYING HEIGHT” and filed on Jul. 23, 2021, which is incorporated herein by reference.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC structures (such as three-dimensional transistors) and processing and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed. For example, device performance (such as device performance degradation associated with various defects) and fabrication cost of field-effect transistors become more challenging when device sizes continue to decrease. Although methods for addressing such a challenge have been generally adequate, they have not been entirely satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a top view of dielectric fin structures with varying height, according to one example of principles described herein.

[0005] FIGS. 2A and 2B is a cross-sectional view of dielectric fin structures with varying height taken along a gate structure, according to one example of principles described herein.

[0006] FIG. 3 is a cross-sectional view of dielectric fin structures with varying height taken along source/drain regions, according to one example of principles described herein.

[0007] FIG. 4 is a top view of dielectric fin structures with varying height in which multiple semiconductor fin structures are placed between dielectric fin structures, according to one example of principles described herein.

[0008] FIG. 5 is a top view of dielectric fin structures with varying height in which some dielectric fin structures are wider than others, according to one example of principles described herein.

[0009] FIG. 6 is a cross-sectional view of dielectric fin structures with varying height and varying widths taken along a gate structure, according to one example of principles described herein.

[0010] FIG. 7 is a cross-sectional view of dielectric fin structures with varying height, and gate all-around fin structures, taken along a gate structure, according to one example of principles described herein.

[0011] FIG. 8 is a flowchart showing an illustrative method for forming dielectric fin structures with varying height, according to one example of principles described herein.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to methods of fabricating field-effect transistors (FETs), such as fin-like FETs (FinFETs), gate-all-around FETs (GAA FETs), and/or other FETs.

[0015] Semiconductor structures often involve semiconductor fin structures that act as channels for transistor devices. As the industry moves towards smaller chips and smaller dimensions, the pitch between adjacent fin structures becomes smaller. This complicates the manufacturing process. To improve the manufacturing process, dummy dielectric lines may be placed between real semiconductor fin structures. This provides a uniform environment in the topography, which improves the manufacturing process for higher layers.

[0016] According to principles described herein, the dielectric fin structures may have lower heights at regions where gate cut features are formed. In particular, gate structures may be formed perpendicular to the semiconductor fin structures and the dielectric fin structures. To cut the gate structures, a patterning process is used to remove a portion of the gate structure. This patterning process leaves an opening that is then filled with a gate end dielectric material. Before the opening is filled, an etching process may partially remove the exposed top surface of the dielectric fin structure. Thus, this portion of the fin structure underneath the gate dielectric is lower than other parts of the dielectric fin structure.

[0017] FIG. 1 is a top view of dielectric fin structures with varying height. According to the present example, the top view shows a plurality of semiconductor fin structures **106** within a Shallow Trench Isolation (STI) region **102**. Between the real semiconductor fin structures **106** are dummy dielectric fin structures **104**. The dummy dielectric structures **104** are illustrated with dotted lines to help visually distinguish them from the real semiconductor fin structures **106**. FIG. 1 also illustrates gate structures **108** extending perpendicular to the semiconductor fin structures and the dielectric fin structures. The gate structures may also be cut by gate-cut dielectric structures **114**. Between the gate structures are source/drain structures **112**. These may be epitaxially grown source/drain structures.

[0018] The gate structures **108**, semiconductor fin structures **106**, and source/drain regions **112** form transistor devices. The transistor devices may form an IC, or a portion thereof, that may comprise static random access memory (SRAM) and/or standard logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as p-type field effect transistors (PFETs), n-type FETs (NFETs), multi-gate FETs such as FinFETs and gate all-around (GAA) FETs, metal-oxide semiconductor field effect transistors (MOSFETs), complementary

metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, other memory cells, and combinations thereof. The gate structures **108**, semiconductor fin structures **106**, and source/drain regions **112** will be discussed in more detail below.

[0019] In one example, the semiconductor fin structures **106** may be fabricated using suitable processes including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the semiconductor fin structures **106** by etching initial epitaxial semiconductor layers of the substrate. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes. The semiconductor fin structures **106** may be doped with proper dopants based on whether they are for n-type or p-type transistors.

[0020] The isolation structure **102** may include silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating material. The isolation structure **102** may be shallow trench isolation (STI) features. In an example, the isolation structure **102** is formed by etching trenches in the substrate as part of the semiconductor fin **106** formation process. The trenches may then be filled with isolating material, followed by a chemical mechanical planarization (CMP) process and/or an etch-back process. The isolation structure **102** may include a multi-layer structure, for example, having one or more thermal oxide liner layers. Then, an STI region **102** may be deposited over the patterned fin structures. The STI region may then be recessed to the desired height, thus exposing the top portions of the fin structures.

[0021] The dielectric fin structures **104** extend parallel with the semiconductor fin structures **106**. The dielectric fin structures may be inserted between active semiconductor fins **106** for improving fin density uniformity and for isolating adjacent semiconductor fins **106** and adjacent gate structures. In some examples, the dielectric fin structures **104** may include multiple dielectric layers. The dielectric fins **104** help reduce coupling capacitance between adjacent gate structures. The dielectric fins **104** also provide high etch resistivity during etching processes, thereby providing high dummy fins for isolating nearby source/drain epitaxial features. Due to the presence of the dielectric fins **104**, the source/drain epitaxial features can be grown to maximum or near-maximum volume, which increases strain to the channel and increases source/drain contact landing area for reduced source/drain contact resistance. The dielectric fin structures may have a width within a range of 5-500 nanometers. The dielectric fin structures **104** may be selected from a group consisting of SiO₂, SiOC, SiOCN, SiCN, carbon content oxide, nitrogen content oxide, or combination. In some examples, the dielectric fin structures may be selected from a nitride base dielectric, metal oxide dielectric, Hf oxide (HfO₂), Ta oxide (Ta₂O₅), Ti oxide (TiO₂), Zr oxide (ZrO₂), Al oxide (Al₂O₃), Y oxide (Y₂O₃), or combination.

[0022] The gate structures **108** may include multiple layers, such as a gate dielectric layer and a gate structure layer. The gate dielectric layer may include silicon oxide (SiO₂), silicon oxynitride (SiON), aluminum silicon oxide (AlSiO), a high-k dielectric material such as hafnium oxide (HfO₂), zirconium oxide (ZrO₂), lanthanum oxide (La₂O₃), titanium oxide (TiO₂), yttrium oxide (Y₂O₃), strontium titanate (SrTiO₃), other suitable metal-oxides, or combinations thereof. The gate dielectric layer may be deposited by chemical oxidation, thermal oxidation, atomic layer deposition (ALD), or other suitable methods. The gate structure layer may include a work function metal layer, a metal fill layer, and other suitable layers

such as barrier layer(s) and capping layer(s). The work function metal layer may be a p-type or an n-type work function layer for the p-type FinFETs and n-type FinFETs, respectively. The p-type work function layer comprises a metal such as titanium nitride (TiN), tantalum nitride (Ta₂N₃), ruthenium (Ru), molybdenum (Mo), tungsten (W), platinum (Pt), or combinations thereof. The n-type work function layer comprises a metal such as titanium (Ti), aluminum (Al), tantalum carbide (TaC), tantalum carbide nitride (TaCN), tantalum silicon nitride (TaSiN), or combinations thereof. The work function metal layer may include a plurality of layers and may be deposited by CVD, physical vapor deposition (PVD), and/or other suitable process. The metal fill layer may include aluminum (Al), tungsten (W), cobalt (Co), copper (Cu), and/or other suitable materials. The metal fill layer may be formed by CVD, PVD, plating, and/or other suitable processes.

[0023] The gate structure may include spacers **110** on both sides. The gate spacers **110** may include one or more dielectric layers having silicon nitride (Si₃N₄), silicon oxide (SiO₂), silicon carbide (SiC), silicon carbide nitride (SiCN), silicon oxynitride (SiON), silicon oxycarbide (SiOC), silicon oxycarbide nitride (SiOCN), a low-k dielectric material, other materials, or a combination thereof. The gate spacers **110** may be formed by one or more methods including chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable methods.

[0024] The source/drain features **112** may include epitaxially grown semiconductor material(s) such as epitaxially grown silicon for n-type FinFETs, and may additionally include one or more n-type dopants, such as phosphorus or arsenic. The source/drain features **112** may also include epitaxially grown semiconductor material(s) such as epitaxially grown silicon germanium for p-type FinFETs, and may additionally include one or more p-type dopants, such as boron or indium. The source/drain features **112** may be formed by a low-pressure CVD (LPCVD) process with a silicon-based precursor, a selective epitaxial growth (SEG) process, a cyclic deposition and etching (CDE) process, or other epitaxial growth processes. The source/drain features **112** may be doped based on the type of transistor to be formed (NMOS or PMOS).

[0025] FIG. 2A is a cross-sectional view of dielectric fin structures with varying height taken along a gate structure. FIG. 2A corresponds to cut A-A of FIG. 1. FIG. 2A illustrates a semiconductor substrate **206**. The semiconductor substrate **206** may be a silicon substrate. The semiconductor substrate may be part of a silicon wafer. Other semiconductor materials are contemplated. The substrate **206** may include an elementary (single element) semiconductor, such as silicon, germanium, and/or other suitable materials; a compound semiconductor, such as silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, and/or other suitable materials; an alloy semiconductor, such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, and/or other suitable materials. The substrate **206** may be a single-layer material having a uniform composition. Alternatively, the substrate **206** may include multiple material layers having similar or different compositions suitable for IC device manufacturing. In one example, the substrate **206** may be a silicon-on-insulator (SOI) substrate having a silicon layer formed on a silicon oxide layer. In another example, the substrate **206** may include a conductive layer, a semiconductor layer, a dielectric layer, other layers, or combinations thereof.

[0026] FIG. 2A also illustrates doped wells **202**, **204**. Doped wells **202** are of P-conductivity type (e.g., doped with p-type impurities such as boron), and are suitable for forming NMOSFETs (e.g., n-type FinFETs). Thus, the transistors formed with the semiconductor fin structures **106** above the p-well may be n-type transistors. Doped well **204** is of N-conductivity type (e.g., doped with n-type impurities such as phosphorous), and is suitable for forming PMOSFETs (e.g., p-type FinFETs). Thus, the transistors formed with the semiconductor fin structures **106** above the n-well may be p-type transistors.

[0027] It can be seen in the cross-sectional view of FIG. 2A that there are two types of dielectric fin structures **104**. The first type of dielectric fin structure **104a** is positioned beneath a gate structure **108**. The second type of dielectric fin structure **104b** is positioned beneath a gate cut feature **114**. While the gate cut feature **114** and the dielectric fin structure **104b** are shown with similar widths in

FIG. 2A, in some cases the gate cut features **114** may be wider than the dielectric fin structures **104**. In some examples, the gate cut features **114** may be narrower than the dielectric fin structures **104**. [0028] In the present example, the first type dielectric fin structure **104a** has a top surface that is higher than the top surface of the second type dielectric fin structure **104b**. The height difference **210** may be within a range of about 5-30 nanometers. Additionally, the top surface of the semiconductor fin structure **106** may be higher than the top surface of the first-type dielectric fin structure **104a**. This height difference **208** may be within a range of about 3-20 nanometers. [0029] FIG. 2B illustrates a cross-sectional view along the dielectric fin structure under the gate electrode. The cross-section corresponds to cut C-C in FIG. 1. As can be seen, in the space between source/drain regions, the dielectric fin structure has a top surface that is lower than the portion where there are gate cut features **114**. The height difference **214** may be within a range of about 3-50 nanometers. [0030] FIG. 3 is a cross-sectional view of dielectric fin structures with varying height taken along source/drain regions FIG. 3 corresponds to cut B-B of FIG. 1. As can be seen, each of the dielectric fin structures **104** has a similar height **304** along this cross-section. The source/drain features **112** may be formed by partially removing portions of the semiconductor fin structures **106** and then replacing those removed portions with an epitaxially grown doped source/drain region. In some examples, an Interlayer Dielectric (ILD) layer **302** may be formed over the source/drain regions **112** and dielectric fin structures **104**. [0031] FIG. 4 is a top view of dielectric fin structures with varying height in which multiple semiconductor fin structures are placed between dielectric fin structures. In other words, while the example in FIG. 1 has only a single semiconductor fin structure positioned between each neighboring pair of dielectric fin structures, the example of FIG. 4 includes more than one (in this case two) semiconductor fin structures **106** between neighboring pairs of dielectric fin structures **104**. [0032] FIG. 5 is a top view of dielectric fin structures with varying height in which some dielectric fin structures are wider than others. In the present example, there are two types of dielectric fin structures. The first type of dielectric fin structures **104** are similar to those described in the example of FIG. 1. The second type of dielectric fin structures **502** are wider than the first type of dielectric fin structures. [0033] FIG. 6 is a cross-sectional view of dielectric fin structures with varying height and varying widths taken along a gate structure. FIG. 6 illustrates a cross-sectional view along cut A-A of FIG. 5. As can be seen the second type dielectric fin structure **502** includes two dielectric layers **602**, **604**. The first dielectric layer **602** may surround the second layer **604**. To form the second-type dielectric line, a trench may be patterned into the STI layer **102** before it is recessed to expose the semiconductor fin structures **106**. Then, the first layer **602** may be conformally deposited along the bottom and sidewalls of that trench. The second layer **604** may then be used to fill the trench. Then, the first layer material **602** may be deposited on top of the second layer **604**. [0034] The gate cut feature **114** may be formed by using a patterning process to form an opening in the gate structure **108**. This opening effectively cuts the gate structure. This opening may be formed using an etching process that partially etches through the first layer **602** of the second type dielectric line **502**. The opening can then be filled with a dielectric material that electrically isolates the separated portions of the gate structure **108**. In one example, the distance **606** between the bottom of the gate cut feature **114** and the top of the dielectric line **502** is within a range of about 5-30 nanometers.

[0035] FIG. 7 is a cross-sectional view of dielectric fin structures with varying height, and gate all-around fin structures, taken along a gate structure. The example of FIG. 7 is similar to that of FIGS. 1-3, but the semiconductor fin structures are fabricated as gate-all-around (GAA) fin structures. In some examples, to form a GAA device, a semiconductor fin may include a total of three to ten alternating layers of semiconductor materials. For example, the first semiconductor material may

be silicon, and the second semiconductor material may be silicon germanium. Either of the first and the second semiconductor materials (or both) may be doped with a suitable dopant, such as a p-type dopant or an n-type dopant, for forming desired FETs. The semiconductor materials and may each be formed by an epitaxial process, such as, for example, a molecular beam epitaxy (MBE) process, a CVD process, and/or other suitable epitaxial growth processes.

[0036] Alternating layers of the semiconductor materials are configured to provide nanowire or nanosheet devices such as GAA FETs, the details of forming which are provided below. GAA FETs have been introduced in effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects. A multi-gate device such as a GAA FET generally includes a gate structure that extends around its channel region (horizontal or vertical), providing access to the channel region on all sides. The GAA FETs are generally compatible with CMOS processes, allowing them to be aggressively scaled down while maintaining gate control and mitigating short-channel effects. Of course, the present disclosure is not limited to forming GAA FETs only and may provide other three-dimensional FETs such as FinFETs.

[0037] In a GAA device, a channel stack is formed by depositing alternating layers of semiconductor material that may be selectively etched. For example, a first type of semiconductor material may be epitaxially grown on a substrate. Then, a second type of semiconductor material may be epitaxially grown on that first layer. The process continues by forming alternating layers of the first and second semiconductor material. Then, the channel stacks may be patterned into fin structures. Each fin may thus be a fin stack of alternating semiconductor layers. Then, an etching process (e.g., a wet etching process) can be used to remove the second semiconductor material while leaving the first semiconductor material substantially intact. The remaining second semiconductor material may thus form a stack of nanowires or nanosheets extending between two active regions. A gate device can then be formed to completely surround each of the nanowires or nanosheets. On each side of the gate device is a source or drain region.

[0038] A lateral etching process to partially remove the sacrificial semiconductor layers where the inner spacers **108** are to be formed. The lateral etching process may be, for example, a wet etching process. The etching process may be designed to be selective so as to remove the sacrificial semiconductor layer without substantially affecting the semiconductor layer. For example, in the case where the sacrificial semiconductor layer is silicon germanium and the semiconductor layer is silicon, then the etching process may be configured to remove silicon germanium without substantially affecting silicon.

[0039] Then, a deposition process is applied to form the inner spacers. Specifically, the inner spacer layer is formed by a conformal deposition process so that the inner spacer layer is formed along sidewalls of the recesses where the source or drain structures are to be formed. The inner spacer layer may be a dielectric material such as SiCN, SiOCN, or SiON.

[0040] An etch back process may then be used to remove portions of the inner spacer layer and to expose the channel layers. The etch back process also removes the inner spacer layer from the floor of the recesses and the top of the workpiece. The remaining portions of the inner spacer layers serve to electrically isolate the portions of the gate structure with the source and drain regions to be formed. In some examples, the remaining inner spacer layer may vary within a range of width between about 4-15 nanometers.

[0041] A wet etching process may then be used to remove the sacrificial layers between the nanostructures. The wet etching process may use an acid-based etchant such as: sulfuric acid (H₂SO₄), perchloric acid (HClO₄), hydroiodic acid (HI), hydrobromic acid (HBr), nitric acid (HNO₃), hydrochloric acid (HCl), acetic acid (CH₃COOH), citric acid (C₆H₈O₇), potassium periodate (KIO₄), tartaric acid (C₄H₆O₆), benzoic acid (C₆H₅COOH), tetrafluoroboric acid (HBF₄), carbonic acid (H₂CO₃), hydrogen cyanide (HCN), nitrous acid (HNO₂), hydrofluoric acid (HF), or phosphoric acid (H₃PO₄). In some examples, an alkaline-based etchant may be used. Such etchants may include but are not limited to ammonium hydroxide (NH₄OH) and potassium

hydroxide (KOH). By removing the sacrificial semiconductor layers, the channel layers thus become nanostructures extending between source and drain structures.

[0042] After the dummy gate structure is removed, a real gate structure is formed. Formation of the real gate device may include a number of steps. For example, a high-k dielectric layer may be deposited so as to surround the channel layers **106**. The high-k dielectric layer may include, for example, aluminum oxide, hafnium oxide, zirconium oxide, hafnium aluminum oxide, or hafnium silicon oxide. Other materials may be used as well. For example, other materials with a dielectric constant greater than 7 may be used.

[0043] In some examples, depending on the type of transistor device being formed, a work function layer may be deposited. Such metal is designed to metal gates the desired properties for ideal functionality. Various examples of a p-type workfunction metal may include, but are not limited to, tungsten carbon nitride (WCN), tantalum nitride (TaN), titanium nitride (TiN), titanium aluminum nitride (TiAlN), tungsten sulfur nitride (TSN), tungsten (W), cobalt (Co), molybdenum (Mo), etc. Various examples of n-type workfunction metals include, but are not limited to, aluminum (Al), titanium aluminum (TiAl), titanium aluminum carbide (TiAlC), titanium aluminum silicon carbide (TiAlSiC), tantalum aluminum silicon carbide (TaAlSiC), and hafnium carbide (HfC). Then, a gate layer is deposited. The gate layer may be a conductive material such as a metal material. In this manner, the gate layer entirely surrounds each of the channel layers.

[0044] FIG. **8** is a flowchart showing an illustrative method for forming dielectric fin structures with varying height. According to the present examples, the method includes a process **802** for forming a first dielectric fin structure and a second dielectric fin structure (e.g., **104**) in parallel on a substrate. The method **800** further includes a process **804** for forming at least one semiconductor fin structure (e.g., **106**) positioned between the first dielectric fin structure and the second dielectric fin structure.

[0045] In one example, the semiconductor structure is formed before the dielectric fin structures. Specifically, the semiconductor fin structures may be formed and then covered with a dielectric material such as an STI material. Before recessing the STI material to expose the upper portions of the fin structures, the STI material may be patterned to form trenches. Those trenches may then be filled with the material to form the dielectric fin structures. This dielectric material may be selectively etchable from the STI material. In this manner, the etching may recess the STI structure without substantially affecting the dielectric fin structures.

[0046] The semiconductor fin structures may be fabricated using suitable processes including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the semiconductor fin structures **106** by etching initial epitaxial semiconductor layers of the substrate. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes. The semiconductor fins may be doped with proper dopants based on whether they are for n-type or p-type transistors.

[0047] Dielectric fin structures may extend parallel with the semiconductor fin structures. The dielectric fin structures may be inserted between active semiconductor fins for improving fin density uniformity and for isolating adjacent semiconductor fins **106** and adjacent gate structures. In some examples, the dielectric fins may include multiple dielectric layers. The dielectric fins help reduce coupling capacitance between adjacent gate structures. The dielectric fins also provide high etch resistivity during etching processes, thereby providing high dummy fins for isolating nearby source/drain epitaxial features. Due to the presence of the dielectric fins, the source/drain epitaxial

features can be grown to maximum or near-maximum volume, which increases strain to the channel and increases source/drain contact landing area for reduced source/drain contact resistance. The dielectric fin structures may have a width within a range of 5-500 nanometers. The dielectric fin structures may be selected from a group consisting of SiO₂, SiOC, SiOCN, SiCN, carbon content oxide, nitrogen content oxide, or combination. In some examples, the dielectric fin structures may be selected from a nitride base dielectric, metal oxide dielectric, Hf oxide (HfO₂), Ta oxide (Ta₂O₅), Ti oxide (TiO₂), Zr oxide (ZrO₂), Al oxide (Al₂O₃), Y oxide (Y₂O₃), or combination.

[0048] The method further includes a process **806** for forming a gate structure (e.g., **108**) over the first dielectric fin structure, the semiconductor fin structure, and the second dielectric fin structure. Forming the gate structure may involve a photolithographic patterning process to form elongated conductive structures that extend perpendicular to the fin structures. The gate structures for a circuit may originally be formed as long lines, which then need to be cut into several smaller gate structures.

[0049] The method **800** further includes a process **808** for removing a portion of the gate structure over the second dielectric fin structure to form an opening to expose a top surface of the second dielectric fin structure. This may be done using a patterning process. Specifically, a photoresist may be deposited onto the workpiece. The photoresist may then be exposed to a light source through a photomask. The photoresist may then be developed so that regions corresponding to where gate cut features are to be formed are exposed. An etching process may then be applied to remove the portion of the gate structure. The opening formed thus separates, or cuts, the gate structure.

[0050] The method **800** further includes a process **810** for etching back a top surface of the second dielectric fin structure through the opening such that the top surface of the second dielectric fin structure within the opening is lower than a top surface of the first dielectric fin structure. In other words, the etching process to remove a portion of the gate structure may continue until a portion of the underlying dielectric fin structure is removed.

[0051] The method **800** further includes a process **812** for filling the opening with a dielectric material to form a gate-cut feature. This may be done using a deposition process such as Chemical Vapor Deposition (CVD), Atomic Layer Deposition (ALD), Plasma Vapor Deposition (PVD), or other suitable deposition techniques. In some examples, a CMP process may be applied afterwards to planarize the surface of the workpiece.

[0052] By using principles described herein, semiconductor device fabrication can be improved. The dummy dielectric fin structures (e.g., **104**) provide for topography uniformity, which benefits the fabrication processes for upper layers (gate, contact, metal interconnect layers). The dielectric fin structures may also help isolate neighboring source/drain regions. This can allow larger dimensions for source/drain regions, which can improve device performance. It can also allow for good performance with single fin transistors, which reduces chip space. The larger source/drain region can also make it easier for contact alignment. The larger area of junction between source/drain and contact also reduces contact resistance.

[0053] According to one example, a semiconductor device includes a semiconductor fin structure extending in a first direction on a substrate and a first dielectric fin structure extending parallel to the fin structure, the first dielectric fin structure being underneath a gate structure extending in a second direction that is perpendicular to the first direction. The device further includes a second dielectric fin structure extending parallel to the fin structure, the second dielectric feature being positioned beneath a gate cut feature. A top surface of the first dielectric fin structure is higher than a top surface of the second dielectric fin structure.

[0054] According to one example, a semiconductor device includes a first dielectric fin structure disposed on a substrate and extending in a first direction, a second dielectric fin structure extending parallel to the first dielectric fin structure, and a semiconductor fin structure extending in the first direction and being positioned between the first dielectric fin structure and the second dielectric fin

structure. A top surface of the first dielectric fin structure is higher than a top surface of the second dielectric fin structure.

[0055] According to one example, a method includes forming a first dielectric fin structure and a second dielectric fin structure in parallel on a substrate, forming at least one semiconductor fin structure positioned between the first dielectric fin structure and the second dielectric fin structure, forming a gate structure over the first dielectric fin structure, the semiconductor fin structure, and the second dielectric fin structure, removing a portion of the gate structure over the second dielectric fin structure to form an opening to expose a top surface of the second dielectric fin structure, etching back a top surface of the second dielectric fin structure through the opening such that the top surface of the second dielectric fin structure within the opening is lower than a top surface of the first dielectric fin structure, and filling the opening with a dielectric material to form a gate-cut feature.

[0056] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device comprising: a semiconductor fin; a first dielectric fin structure and a second dielectric fin structure, wherein the semiconductor fin is between the first dielectric fin structure and the second dielectric fin structure; a shallow trench isolation region disposed between the first dielectric fin structure and the semiconductor fin and between the second dielectric fin structure and the semiconductor fin, wherein the shallow trench isolation region underlies the first dielectric fin structure and the second dielectric fin structure; and a gate structure overlying the first dielectric fin structure and the semiconductor fin; and a gate cut feature overlying the second dielectric fin structure, wherein: at a location underlying the gate structure, the first dielectric fin structure has a first height from a bottom surface of the first dielectric fin structure to a top surface of the first dielectric fin structure, and at a location underlying the gate cut feature, the second dielectric fin structure has a second height from a bottom surface of the second dielectric fin structure to a top surface of the second dielectric fin structure, and the second height is different than the first height.
